Rice University

Uncovering Strategic Archetypes: A Behavioral Clustering Analysis of NASCAR Cup Series Drivers and Application to Race Prediction

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SMGT 490

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May 3, 2026

Abstract

This study applies unsupervised machine learning to identify behavioral archetypes among NASCAR Cup Series drivers, shifting the analytical focus from outcome prediction to process-oriented style classification. Using race-level data from 2017 to 2024, behavioral features were engineered relative to the field within each race and clustered independently within four track types: short tracks, intermediates, superspeedways, and road courses. Twelve distinct behavioral archetypes emerged across track types, and 62.5% of drivers exhibited different behavioral styles depending on track type, demonstrating genuine racing adaptation rather than a single fixed identity. A Random Forest regression model incorporating behavioral archetype as a feature predicted race finishing positions with an RMSE of 10.47 and championship standings with a Spearman correlation of 0.72–0.83 across holdout seasons. Behavioral archetype contributed a 10.1% increase in mean squared error when removed from the model, confirming it adds meaningful predictive signal beyond starting position and historical performance. An interactive web application was developed to make these findings accessible to teams and analysts.

1. Introduction

NASCAR is one of the premier motorsport series in North America, drawing approximately 3.2 million viewers per race weekend and generating significant strategic complexity through its stage-based points system, track type variability, and 36-race championship season. Despite this complexity, the academic analytics literature on NASCAR has remained almost entirely outcome-focused, prioritizing the prediction of finishing positions over the understanding of how drivers actually compete.

This gap stands in stark contrast to the trajectory of analytics in other major sports. Basketball and soccer analytics have moved decisively toward process-oriented, style-discovery frameworks, clustering players by behavioral patterns rather than production statistics, and identifying stylistic identities that inform opponent preparation, player development, and in-game tactical decisions (Akhanli & Hennig, 2023; Ajinaja et al., 2022). NASCAR generates analogous process data through lap timing, position changes, stage results, and finishing records, yet this data has almost exclusively been applied to outcome prediction.

This study closes that gap. The central research question is: *Can NASCAR Cup Series drivers be grouped into distinct behavioral archetypes based on their in-race tendencies, and do these archetypes help predict performance across different track types?*

The research makes two primary contributions. First, it introduces a track-type-specific behavioral clustering framework that identifies twelve distinct driver archetypes — three per track type, reflecting genuine variation in how drivers approach different racing environments. Second, it demonstrates that behavioral archetype membership carries statistically significant predictive value for race outcomes, contributing meaningful signal to a Random Forest race prediction model beyond traditional performance metrics.

2. Literature Review

2.1 Outcome-Focused Models in NASCAR Analytics

Academic research on NASCAR has been dominated by econometric and predictive modeling centered on the question of what factors determine who wins. Allender (2008) established the paradigmatic approach by developing a regression model predicting finishing order using driver experience, starting position, and career earnings. Her findings highlighted the statistical significance of veteran experience as a predictor of race outcomes. Subsequent work extended this framework through tournament theory applied to prize structure optimization (Von Allmen, 2001) and Data Envelopment Analysis for team efficiency measurement.

These studies share a fundamental orientation: the driver is treated as a black box that converts inputs like skill, resources, equipment into an output, finishing position. While this approach satisfies predictive needs, Alamar (2013) identifies its core limitation: outcome-focused analytics provides little actionable intelligence for in-race decision-making. A crew chief cannot learn from a regression model how an experienced driver handles tire degradation, executes passes, or manages stage-point strategy. The outcome-focused paradigm meets a forecasting need while failing the demand for process-oriented, practically useful intelligence.

2.2 The Methodological Shift in Other Sports

Analytics in dynamic team sports has moved toward a more sophisticated goal: understanding how athletes perform rather than just predicting who wins. Two methodological developments are particularly relevant.

Decroos et al. (2020) introduced the VAEP (Valuing Actions by Estimating Probabilities) framework for soccer, which assigns value to every on-ball action based on its contribution to scoring and conceding probability. This represents a fundamental shift from goal-counting to decision-quality assessment: measuring the behavioral process rather than the aggregate outcome.

Akhanli and Hennig (2023), publishing in the Journal of Quantitative Analysis in Sports, applied this process-focused orientation to player clustering. Using in-game action measurements across European football players, they applied K-Means clustering with aggregated validity indexes to identify archetypes defined by behavioral patterns, like "ball-winning defenders" and "creative midfielders", classified by how they affect the game rather than their statistical production. Ajinaja et al. (2022) similarly employed K-Means and hierarchical clustering to identify NBA player roles from statistical profiles, distinguishing defensive anchors from perimeter scorers.

This corpus of work establishes a clear methodological template: transform low-level event data into behavioral features, then apply unsupervised learning to identify organic groupings. The fundamental question shifts from "who is best?" to "what is their style?"

2.3 The Identified Research Gap

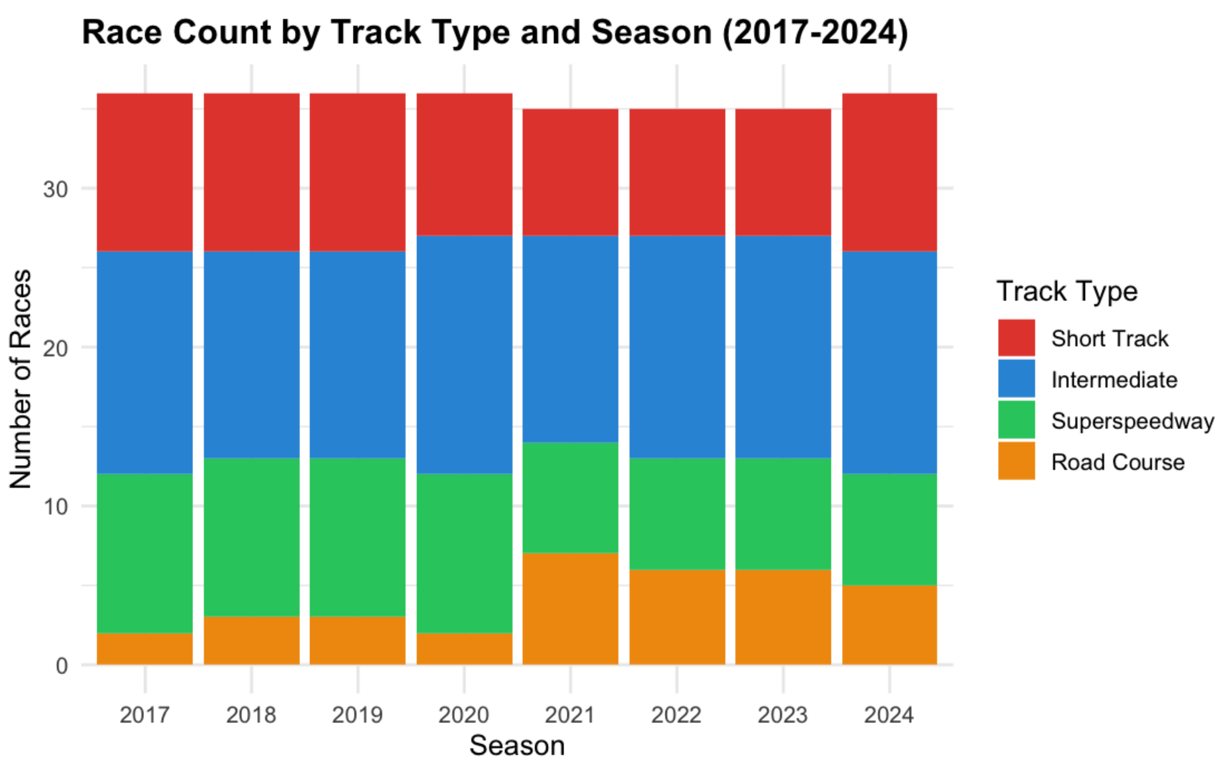
The comparison of these two analytical traditions reveals a clear gap. NASCAR generates rich process data structurally similar to the pass-by-pass and shot-by-shot data used in basketball and soccer analytics: lap timings, position changes, pit cycles, stage results. Yet the process-focused paradigm has not been applied to motorsports. No systematic framework exists for classifying NASCAR drivers into behavioral archetypes based on how they make decisions during a race.

This study fills that gap directly, applying the methodological steps demonstrated in the literature: motivated by the demand for actionable intelligence (Alamar, 2013), developing behavioral metrics from in-race data (inspired by Decroos et al., 2020), and identifying organic driver groupings through clustering analysis (following Akhanli & Hennig, 2023; Ajinaja et al., 2022).

3. Data

3.1 Source and Study Period

Data was sourced from the `nascaR.data` R package (Grealis, 2024), which provides historical race-by-race results for the NASCAR Cup Series from 1949 to present, scraped with permission from DriverAverages.com. The study period covers the 2017 through 2024 seasons, corresponding to the stage racing era introduced in 2017. This era is analytically distinct from prior periods because stage points create genuine within-race strategic decisions: drivers must actively choose whether to race for stage results or conserve for the finish, making behavioral differentiation more observable in the data.



3.2 Sample

The dataset was filtered to full-time drivers, defined as drivers who averaged 30 or more races per season across their seasons in the study period. This threshold captures drivers with a consistent presence in the championship, ensuring behavioral profiles are based on sufficient race samples. Drivers competing part-time, whether retired legends running select events or road course specialists like Shane Van Gisbergen, were excluded, though this represents an acknowledged limitation discussed in Section 7. The final sample comprises 32 drivers across 6,992 driver-race entries.

Dirt races (n=3 in the study period, all Bristol Motor Speedway dirt experiments in 2021 and 2022) were excluded as behaviorally distinct from the four primary track types and negligible in count.

3.3 Track Classification

Tracks were classified into four types using a combination of surface type and track length. Road surface tracks were classified as Road Courses. Paved tracks were classified as Short Tracks (≤1.0 mile), Intermediates (1.0–2.0 miles), or Superspeedways (≥2.0 miles). Two tracks with missing metadata, Autodromo Hermanos Rodriguez (a road course) and EchoPark Speedway (a 1.333-mile intermediate), were manually classified based on known track characteristics.

The resulting study dataset contained 110 intermediate races, 73 short track races, 68 superspeedway races, and 34 road course races across the eight-season period.

4. Methodology

4.1 Feature Engineering

The core methodological challenge in behavioral analytics with race-level data is that behavioral signals and outcome signals are difficult to fully separate. A driver who gains positions relative to the field may be skilled, aggressive, or both. To maximize the behavioral signal while minimizing outcome contamination, all features were computed relative to the field within each race. This ensures that measurements reflect how a driver performed compared to their competitors on a given day, rather than against all-time baselines that would conflate behavioral tendencies with equipment quality and overall skill level.

Four behavioral features were engineered:

Relative Positions Gained captures overtaking behavior. For each driver in each race, positions gained (start minus finish) is computed, then the field average positions gained for that race is subtracted. A positive value indicates the driver moved through the field more than typical for that race; a negative value indicates the opposite. This field-normalization ensures that a driver starting last — who has structural advantages in positions gained — is evaluated against the same baseline as a driver starting at the front.

Relative Stage Points captures stage-hunting strategy. Driver stage points (sum of S1, S2, S3) minus the field average stage points for that race reflects whether a driver actively competed for stage results beyond what their general competitiveness would predict.

Relative Finish Variance captures consistency versus boom-or-bust racing style. The variance of a driver's field-relative finish positions across all races at a given track type reflects whether they produce predictable results or oscillate between strong performances and poor ones.

DNF Rate captures risk tolerance and reliability. The proportion of races in which the driver did not finish reflects a combination of aggressive driving choices, mechanical reliability, and crash involvement.

To further address manufacturer quality confounding, make-adjusted versions of position gained and stage points were computed by subtracting the manufacturer average for each metric within each race. This ensures that a driver in elite equipment does not appear behaviorally more aggressive simply because their car allows execution of aggressive strategies unavailable to others.

4.2 Within-Track-Type Clustering

A critical methodological decision in this study is to cluster drivers *within* each track type rather than producing a single overall behavioral archetype per driver. This decision is motivated by the observation that driver behavior genuinely adapts to racing context. A driver may be an aggressive stage hunter on intermediate tracks while adopting a conservative survival strategy at superspeedways. Producing a single overall archetype would mask these contextual shifts and produce groupings that primarily reflect overall skill level, precisely the outcome-focused orientation this study seeks to move beyond.

For each of the four track types, driver behavioral profiles were computed by aggregating the four relative features across all races at that track type. A minimum of 10 races per driver-track-type combination was required for inclusion, resulting in the exclusion of five thin combinations, four road course profiles and one short track profile, belonging to drivers with limited appearances at those track types due to partial seasons or retirement during the study period.

K-Means clustering with k=3 was applied independently within each track type. The choice of K-Means was motivated by its interpretability, its production of hard non-overlapping cluster assignments well-suited to discrete archetype labeling, and its precedent in the sports analytics literature (Akhanli & Hennig, 2023; Ajinaja et al., 2022). The optimal number of clusters was determined using the elbow method and silhouette analysis. Cluster stability was assessed using bootstrap resampling with 500 iterations via the `fpc` package, measuring Jaccard similarity between cluster solutions across bootstrap samples. Scores above 0.75 indicate stable clusters, 0.60–0.75 borderline, and below 0.60 unstable.

4.3 Cluster Interpretation and Naming

After clustering, each cluster was interpreted by examining its centroid values across the four behavioral features. Clusters were then assigned descriptive archetype names reflecting their dominant behavioral pattern. This interpretation process followed the approach of Akhanli and Hennig (2023), who similarly named clusters based on centroid feature values and validated naming through qualitative cross-reference with domain knowledge.

4.4 Race Outcome Prediction Model

To demonstrate the applied value of behavioral archetypes, a two-stage Random Forest regression model was developed to predict race finishing positions.

The enhanced feature set included starting position, track type, behavioral archetype at that track type, historical relative finish at that track type, recent form (rolling average finish over the last five races, computed using a lag to prevent data leakage), track-specific finishing history (cumulative mean of prior finishes at the exact track, also lagged), and manufacturer as a categorical variable.

The model was trained on seasons 2017 through 2022 and evaluated on the holdout period of 2023 and 2024, a temporal split that reflects realistic deployment conditions where future races are predicted using only past data. Model performance was evaluated using RMSE and MAE on individual race predictions, as well as Spearman rank correlation between predicted and actual championship standings across the two holdout seasons.

Stage point predictions for Stages 1 and 2 were generated using separate Random Forest models with identical feature sets, enabling full points-based championship standing predictions using the official NASCAR points structure.

5. Results

5.1 Behavioral Archetypes

The clustering procedure identified three distinct behavioral archetypes within each of the four track types, producing twelve archetypes in total. Table 1 summarizes the centroid values and average relative finishing position for each archetype.

Short Track Archetypes:

Stage Point Collectors (n=17) represent the largest and most successful short track archetype. These drivers lose positions relative to the field on average, suggesting they begin races conservatively, but accumulate stage points aggressively (avg relative stage pts: +1.58 vs field) with a very low DNF rate of 6.5%. Their average relative finish of -3.09 positions identifies them as the field's strongest short track performers. Elite drivers including Kevin Harvick, Denny Hamlin, Martin Truex Jr., Chase Elliott, and Kyle Larson populate this archetype.

Late Race Movers (n=8) gain positions relative to the field (+2.04) while avoiding stage points and maintaining low DNF rates. Their average relative finish of +3.56 suggests they sacrifice early race positioning, potentially conserving tires, for late-race charges that do not fully overcome their stage point deficit.

Crashers (n=6) are defined primarily by their elevated DNF rate of 13.9%, the highest of any short track archetype. Their average relative finish of +5.79 positions reflects the cost of attrition-prone aggressive racing on tight short tracks.

Intermediate Track Archetypes:

Stage Hunting Chargers (n=15) gain positions relative to the field and accumulate stage points aggressively, producing an average relative finish of -2.84, the best among intermediate archetypes. This archetype captures the dominant mid-race racing style of elite oval drivers.

Conservative Managers (n=13) lose positions and avoid stage points, but produce very consistent relative finishes. Their average of +3.67 reflects a risk-averse approach that prioritizes reliability over peak performance.

Position Climbers (n=4) gain the most positions of any intermediate archetype (+2.58 relative to field) while completely ignoring stage points, with an average relative finish of +4.97. This pattern is consistent with backmarker drivers who start near the rear and move through the field without the speed to threaten for stage results.

Road Course Archetypes:

Road Course Racers (n=12) gain positions and hunt stage points, producing an average relative finish of -2.04. This archetype captures drivers who are genuinely competitive on road courses, actively engaged in the race rather than simply managing attrition.

Points Survivors (n=11) lose positions and avoid stage points, but maintain very low DNF rates (3.9%) and an average relative finish of -0.58. This archetype represents oval specialists who survive road courses by racing conservatively, accepting modest results without crashing out.

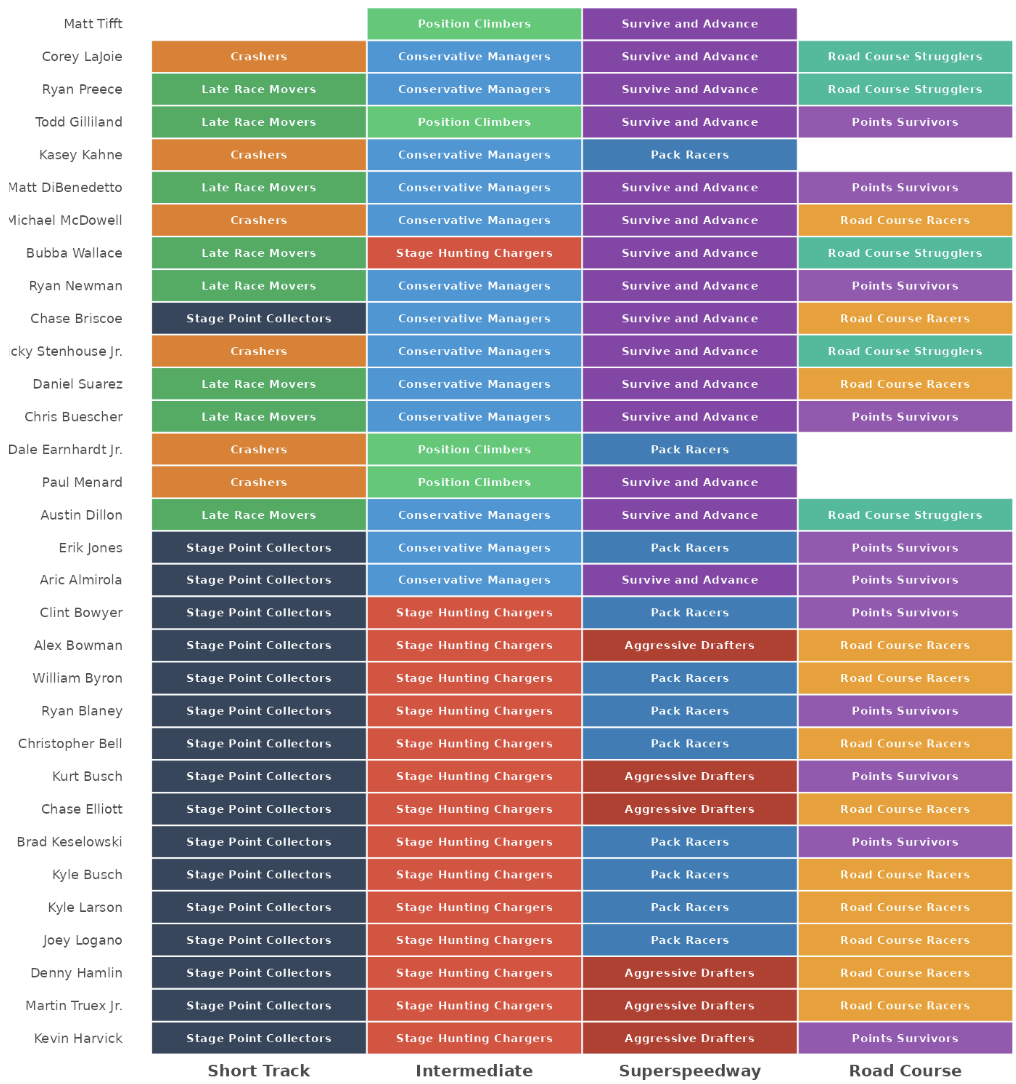
Road Course Strugglers (n=5) have the highest DNF rate of any road course archetype at 17.7%, producing an average relative finish of +5.44. These drivers are most exposed to road course attrition.

Superspeedway Archetypes:

Aggressive Drafters (n=6) gain positions and hunt stage points with a high DNF rate of 15.6%, producing the best average relative finish at superspeedways (-2.65). This archetype captures drivers who push aggressively in draft situations, maximizing their upside at the cost of elevated crash risk.

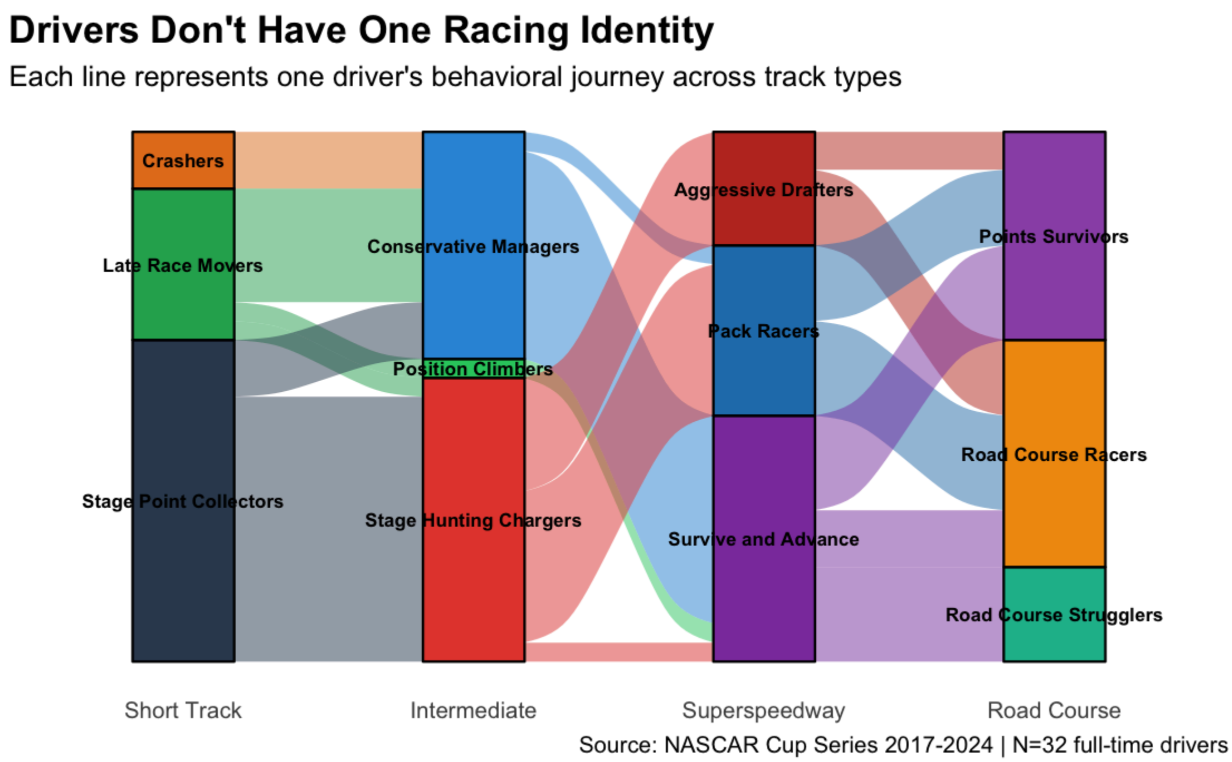
Pack Racers (n=11) also gain positions and hunt stage points, but with dramatically higher finish variance (139.6) and DNF rate (26.2%), the highest of any archetype across any track type. Their average relative finish near zero reflects the volatility of their racing approach, trading high DNF exposure for occasional strong finishes.

Survive and Advance (n=15) is the largest superspeedway archetype. These drivers lose positions, avoid stage points, and produce modest average results (+2.37). This strategy accepts lower finishing positions in exchange for reduced crash exposure at tracks where attrition dramatically reshapes results.

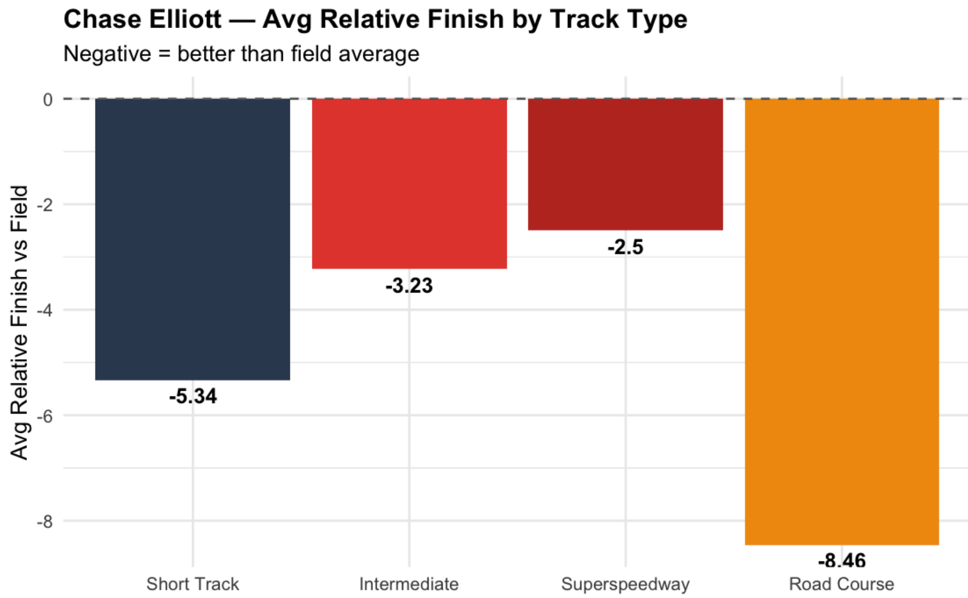


5.2 Behavioral Adaptation Across Track Types

A central finding of this study is that most NASCAR Cup Series drivers do not exhibit a single fixed behavioral style across all track types. Of 32 full-time drivers analyzed, 8 (25%) shifted across three distinct behavioral styles depending on track type, 20 (62.5%) shifted between two styles, and only 1 (Matt Tifft, with limited road course and short track data) maintained a single style across all track types.



Chase Elliott provides the clearest illustration of behavioral adaptation. On short tracks he is classified as a Stage Point Collector: methodical, low DNF, and hunting stage points. On intermediates he shifts to Stage Hunting Charger, actively moving through the field and pursuing stages. On superspeedways he becomes an Aggressive Drafter, pushing hard in pack situations. On road courses he is classified as a Road Course Racer, his statistically strongest track type with an average relative finish of -8.46 positions: the most extreme track-type specialization in the dataset.



Ricky Stenhouse Jr. represents a different kind of adaptation — shifting across three archetypes with his superspeedway behavior most distinctive, consistent with his known restrictor plate racing specialization and multiple Daytona and Talladega victories.

5.3 Statistical Validation

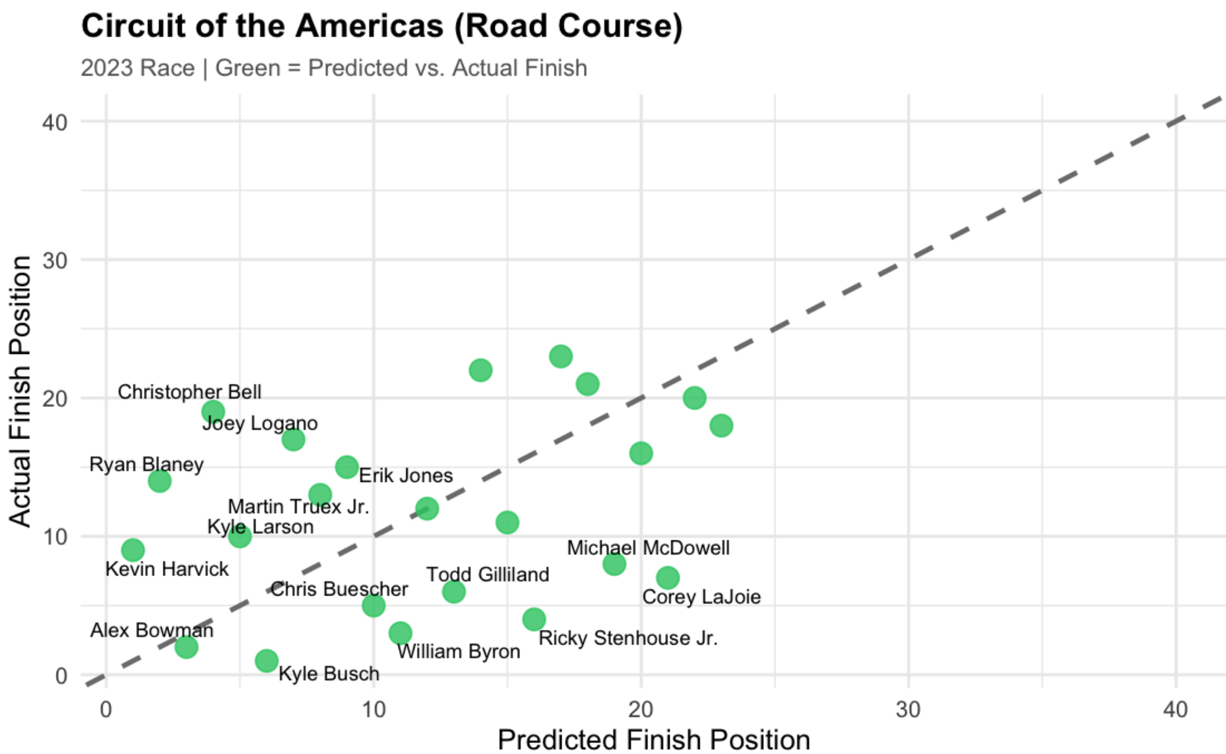
One-way ANOVA tests confirmed that behavioral archetype membership was significantly associated with finishing position across all four track types, with all p-values below 0.001 (Intermediate: $F=136.4$, $p<10^{-82}$; Short Track: $F=137.9$, $p<10^{-80}$; Road Course: $F=25.3$, $p<10^{-15}$; Superspeedway: $F=22.2$, $p<10^{-13}$). Tukey post-hoc tests identified significant pairwise differences between archetype pairs within each track type.

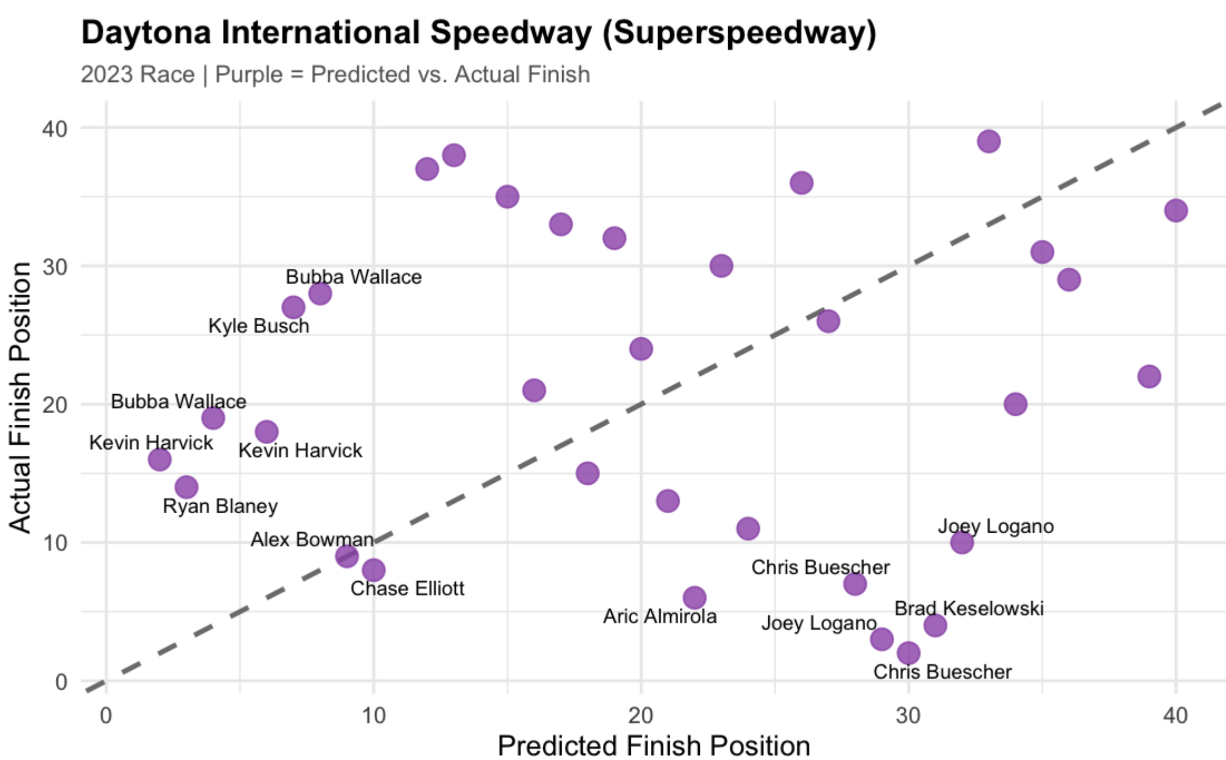
The strength of archetype differentiation varied by track type. Short tracks and intermediates showed the largest F-statistics, indicating archetype membership is most predictive of performance on these track types. Superspeedways showed the smallest F-statistic, reflecting the well-documented "superspeedway equalizer" effect in which pack racing and attrition compress competitive differences between drivers, flattening performance gaps between behavioral archetypes.

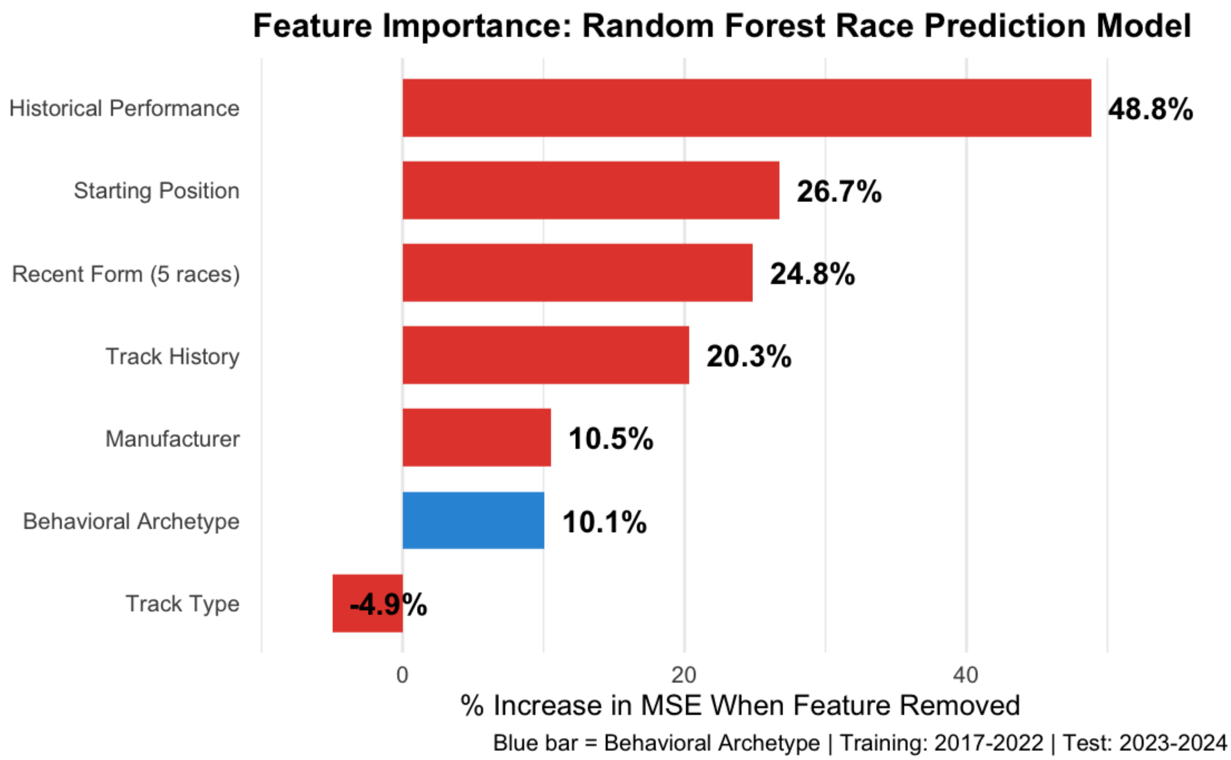
5.4 Race Outcome Prediction

The enhanced Random Forest model achieved an RMSE of 10.47 and MAE of 8.68 on the 2023–2024 holdout test set of 71 races, representing improvement over the baseline model without recent form, track history, and manufacturer features (RMSE 10.83, MAE 8.93).

Feature importance analysis revealed the relative contribution of each predictor. Historical relative finish at the track type was most important (48.8% increase in MSE if removed), followed by starting position (26.7%), recent form over the last five races (24.9%), track-specific finishing history (20.3%), manufacturer (10.5%), and behavioral archetype (10.1%). Track type alone showed slightly negative importance (-4.9%), indicating that once behavioral archetype — which already encodes track-type-specific behavioral tendencies — is included in the model, the raw track type variable adds little additional information.


Circuit of the Americas (Road Course)
2023 Race | Green = Predicted vs. Actual Finish
Actual Finish Position
Predicted Finish Position
Christopher Bell
Joey Logano
Ryan Blaney
Erik Jones
Martin Truex Jr.
Kyle Larson
Kevin Harvick
Michael McDowell
Todd Gilliland
Chris Buescher
Corey LaJoie
Alex Bowman
Ricky Stenhouse Jr.
William Byron
Kyle Busch


Daytona International Speedway (Superspeedway)
2023 Race | Purple = Predicted vs. Actual Finish
Actual Finish Position
Predicted Finish Position
Bubba Wallace
Kyle Busch
Bubba Wallace
Kevin Harvick
Kevin Harvick
Ryan Blaney
Alex Bowman
Chase Elliott
Joey Logano
Chris Buescher
Brad Keselowski
Aric Almirola
Joey Logano
Chris Buescher



Evaluated across all 71 test races, the model correctly identified an average of 38.6% of top-5 finishers per race (versus 12.5% expected by random chance with 32 drivers) and 56.8% of top-10 finishers (versus 25.0% by chance). Applied to championship standing prediction, the model achieved Spearman rank correlations of 0.828 in 2023 and 0.721 in 2024, with average standing errors of 3.6 and 3.8 positions respectively.

6. Web Application

An interactive R Shiny web application was developed to make the findings of this research accessible to teams, analysts, and fans. The application is publicly available at:

https://savanpatel.shinyapps.io/nascar-archetypes/

The application provides three primary features. The Race Preview tab allows users to select any NASCAR Cup Series track and receive a predicted finishing order for all 32 drivers in the study, with color-coded archetype labels and historical archetype performance context for that track type. The Driver Explorer tab provides individual driver profiles including their archetype classification at each track type, average relative finish by track type, and a behavioral radar chart normalized across the full driver sample. The Archetype Browser tab presents the full driver-by-track-type archetype heatmap with filtering by track type and descriptive summaries of each archetype's behavioral characteristics.

7. Limitations

This study acknowledges several methodological limitations that future research should address.

Data granularity represents the most significant constraint. Race-level summary data limits the precision of behavioral measurement. Lap-by-lap position tracking, real-time pit stop timing, and individual passing attempt data, available through sources like the Sportradar NASCAR API, would enable more direct observation of behavioral decisions rather than their aggregated outcomes. The behavioral features used here capture genuine tendencies but cannot fully decouple style from equipment quality and overall driver skill.

Sample size affects cluster stability. With 32 full-time drivers, bootstrap Jaccard stability scores ranged from 0.61 to 0.89 depending on cluster and method. Hierarchical clustering (Ward's method) produced more stable solutions than K-Means, and the two methods agreed perfectly on cluster membership, a positive validation finding, but the small driver sample means results should be interpreted with appropriate caution about generalizability.

Full-time driver filter excludes part-time specialists. Shane Van Gisbergen is the most prominent example: a multiple road course winner in the study period who would almost certainly appear as an extreme outlier in the Road Course Racer archetype. Future work could lower the minimum race threshold specifically for track-type-specific specialists.

Prediction model compression is an inherent limitation of Random Forest regression. Predicted finish positions ranged from approximately 9 to 28 across all predictions, substantially narrower than actual finish positions that span 1 to 40. This compression toward the mean reflects the ensemble averaging property of Random Forests and would be addressed in future work through distributional prediction methods or quantile regression forests.

Manufacturer confounding remains partial despite make-adjustment. Manufacturer competitiveness shifts substantially year to year, like Toyota's dominance in certain periods and Ford's resurgence in others, and the current adjustment captures within-race manufacturer effects but not longer-term equipment quality trends.

8. Conclusion

This study introduces a track-type-specific behavioral clustering framework for NASCAR Cup Series drivers, identifying twelve distinct archetypes that describe how drivers approach different racing environments rather than simply how well they perform. The central finding is that NASCAR drivers are not behaviorally static: most exhibit two to three distinct racing styles depending on whether they are at a short track, intermediate, superspeedway, or road course. This behavioral adaptation is measurable, statistically significant, and predictively useful.

The contribution of behavioral archetype to race outcome prediction, a 10.1% increase in mean squared error when removed from the model, demonstrates that knowing how a driver races at a given track type adds meaningful predictive signal beyond knowing where they start and how they have historically finished. This confirms the core thesis: behavioral style is not simply a proxy for skill, but an analytically meaningful dimension that carries independent information about race outcomes.

For practitioners, the archetype framework offers actionable intelligence that outcome-focused models cannot provide. Understanding that a driver is classified as a Stage Point Collector on short tracks versus a Crasher, with their associated DNF rates and relative performance profiles, informs race strategy preparation, driver evaluation, and opponent modeling in ways that a finishing position average cannot. The interactive web application makes this intelligence accessible without requiring direct engagement with the underlying statistical models.

Future work should prioritize lap-level data integration, which would enable more precise behavioral measurement and allow this framework to be applied in real-time during races for live strategy adjustments. Expanding the driver sample to include part-time track specialists and applying the framework to the O'Reilly Auto Parts Series and Craftsman Truck Series would further validate the generalizability of behavioral archetypes as an analytical lens in motorsports.

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awareness are not observed. From the viewpoint of data science, the COMPAS case shows that fairness, transparency, and accountability are not peripheral values but rather foundational design requirements.

Reduced to their essence, the COMPAS controversy represents a failure to uphold each of the three ethical principles enunciated in the Belmont Report: respect for persons, beneficence, and justice. The principle of respect requires that individuals affected by data systems maintain their autonomy and understand how decisions about them are made. However, according to Vaccaro (2019), COMPAS operates as a proprietary "black box," where neither judges nor defendants can interpret or contest the algorithm's reasoning. This lack of transparency constitutes a violation of respect owed to the individuals who are subject to algorithmic classification, especially when those classifications determine liberty or incarceration. Similarly, beneficence-the obligation to minimize harm and maximize benefit-was breached by COMPAS's racially uneven predictive power. According to Analysis of Racial Bias in Northpointe's COMPAS Algorithm (2019), logistic regression models using only age and prior offenses predicted recidivism more accurately and more justly than COMPAS itself. Rather than mitigating human bias, the system amplified it. Finally, the principle of justice-assuring fair distribution of risk and benefit-was breached when African American defendants were systematically assigned higher risk scores than white defendants with comparable criminal histories. In ethical terms, the model's deployment failed the very standards of human subject research that inspired modern data ethics frameworks.

For data scientists, the COMPAS case also reflects the need to move beyond "ethics washing" toward genuine algorithmic accountability. Corporate environments often prioritize profitability and efficiency, creating a gap between ethical ideals and operational incentives. The COMPAS system, developed by a private firm and sold to state governments, exemplifies this tension. Ethical review processes comparable to Institutional Review Boards in academic research were absent, which allowed the company to deploy a high-stakes predictive model without any kind of external auditing. A principlist approach-where broad ethical principles are interpreted and applied through deliberate institutional mechanisms-would have required Northpointe to assess the potential for disparate impact and show evidence of fairness before use. Algorithmic auditing, now increasingly recognized as a best practice, might have surfaced COMPAS's racial disparities and mitigated harm before it reached the real world. The absence of such oversight reveals how "power without ethics," as discussed in class, lets data systems amass enormous power without accountability.

A virtue ethics perspective adds a further dimension to the data scientist's role in cases such as COMPAS. Ethical practice in data science is not just a matter of rules or compliance standards-it involves the professional virtues of integrity, humility, and social awareness. The development of COMPAS shows what happens when those virtues are ignored: the model's designers failed to question how historical data from a racially biased justice system would condition predictions, and failed to communicate uncertainty or limitations to end users. A virtuous data scientist understands that technical choices-such as which variables to include or exclude-involve moral decisions with the potential to entrench or contest systemic inequities. Embedding these virtues within professional culture requires designing not only for performance but also for human dignity.

Finally, ethical consideration of the COMPAS algorithm calls for reflection from a utilitarian perspective based on outcomes. If an algorithm's purpose is to provide maximum benefit to society by minimizing crime and efficiently determining sentences, then it also needs to take into account the harms caused by false predictions and unfair incarceration. The harms reported in both Vaccaro (2019) and Analysis of Racial Bias in Northpointe’s COMPAS Algorithm (2019), higher rates of false positives among Black defendants and psychological overdependence upon algorithmic scores—represent some of the ways in which the use of COMPAS has fallen short of creating the greatest good for the most people. Beneficence duly requires weighing utility against equity and prioritizing models that achieve efficiency without sacrificing fairness.

In sum, ethical concerns within the COMPAS algorithm demonstrate that data science cannot be dissociated from moral responsibility. Its lack of transparency, reinforcement of racial disparities, and cognitive influence on human decision-makers-all illustrate the dangers of powerful tools applied without principled oversight. For data scientists, the lesson is clear: ethics must not be treated as post-hoc evaluation or sets of constraints but must form part of design considerations. Embedding principlism, algorithm auditing, and fairness-by-design approaches into the development of systems is not just good practice-it is the only way to ensure predictive systems serve justice rather than distort it.

Societal Impacts

With the heavily relied use of RAIs in the judicial system like COMPAS, this sets precedence for the use of algorithms that extend beyond the courts, especially on policymaking and governance. One example is feedback loops in policy design. If COMPAS data is aggregated and later used in criminal justice statistics, it may artificially inflate reoffending risk among minority populations, which can cause a ripple effect on policymakers supporting harsher crime policies in minority neighborhoods. Another example could be distortion of resource allocation, where if COMPAS statistics signal certain communities have higher criminal risk, this could cause funding shifts from rehabilitation or social support towards enforcement, even when the underlying issue is socio-economic inequality.

Implementation of algorithms like COMPAS that are not completely fair in all metrics can cause erosion of trust in the justice system, but also in the technology. When people learn that algorithms are influencing criminal sentencing in a skewed manner, confidence in algorithmic systems would decline. This skepticism can spread beyond courts to other areas where algorithmic systems are being implemented, such as recent developments in AI-powered search engines. However, more likely, the benefits of data science in governance or policymaking would be resisted. Furthermore, marginalized communities that have been disproportionately harmed by these systems would grow even more distrustful of the governments that deploy them, widening the gap between the citizens and the state. To fully realize the future benefits of data science and algorithmic systems, we must build trust in the systems that are already implemented, like the COMPAS algorithm.

Finally, the diffusion of such COMPAS-like systems has larger implications for data governance and digital accountability across society. When obscure algorithms make consequential decisions, they blur lines of authority between human and automated, with questions about who bears responsibility for harm. This diffusion of accountability can normalize a culture of technological determinism-where outcomes are accepted as "data-driven" rather than interrogated for fairness or context. Over time, this desensitizes both policymakers and the public to structural bias and embeds inequity into the fabric of governance under the guise of objectivity. Instead, society should demand transparent auditing standards, algorithmic impact assessments, and participatory oversight mechanisms that give affected communities a voice in how such systems are designed and deployed.

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